

Antoine Lopez

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PROFILE

Performance-focused **Software Engineer** with a strong foundation in low-level systems programming, distributed tooling, and backend architectures. Proficient in Python, C, Rust, and Scala, with active contributions to high-impact open-source developer ecosystems.

EXPERIENCE

AI Scientist Intern

February 2026 – August 2026

ALTEN

Sèvres, France

- Developed an end-to-end framework for predictive system-failure modeling in complex Web3 financial networks (JEPA)
- Engineered an Ego-temporal network pipeline using PyTorch Geometric to capture multi-scale local node dynamics and neighborhood evolution (GCP BigQuery)
- Implemented a JEPA to predict future latent representations of network states (Python, PyTorch)
- Managed the end-to-end research pipeline, including formalizing the theoretical hypothesis, designing custom data ingestion for temporal graphs, and validating model robustness against historical system shocks.

Data Engineer Intern

May 2025 – August 2025

Société Générale

Val de Fontenay, France

- Designed and assessed software testing strategies in a complex environment
- Implemented new testing strategies tailored to requirements (Scala, Java, Spark, Apache Kafka)
- Developed a testing application for cloud data lake ingestion, including APIs, S3 buckets, and data processing pipelines (Python, REST API, AWS S3)

Software Developer Intern

January 2024 – February 2024

VINCI Energies — Cloud, Communication and Services

La Défense, France

- Developed a chatbot for client ticket management using Llama 2 (Python, Docker, Linux VM)

Junior Engineering Office Technician Intern

December 2022 – February 2023

VINCI Energies — AXIANS

Palaiseau, France

- Wrote procedures and processed Work Declarations and Notices of Intent to Commence Work
- Built an Excel database for managing customer contracts (Excel, VBA)

EDUCATION

ECE Paris

September 2021 – August 2026

Master's Degree in Engineering - Big Data & Artificial Intelligence | Cloud Option

Paris, France

Ajou University

August 2023 – January 2024

University Exchange Semester — Computer Science

Suwon, South Korea

French Baccalauréat | *Maths, Physics-Chemistry, Computer Science*

2021

PROJECTS

Transformer from Scratch in C | *C, Low-Level Optimization, LLM*

- Implemented an autoregressive Transformer purely in C with manual tensor and memory management.

Open Source Contribution: uv | *Rust, Systems Programming, Astral*

- Contributed codebase enhancements and fixes to Astral's high-performance Python package manager.

Open Source Contribution: scikit-lego | *Python, Machine Learning, Scikit-Learn*

- Contributed custom transformers and utility estimators to the scikit-lego library.

Tech Watch Agent | *Agentic AI, Python, OpenRouter API*

- Created an autonomous agent to scrape, analyze, and synthesize daily AI research updates via OpenRouter.

TECHNICAL SKILLS

Languages: C, Rust, Python, Scala, Java, SQL

Systems & Infrastructure: Linux, Docker, Git/GitHub, CI/CD

Open Source & Tooling: UV (Astral), Scikit-Lego, API Integration, Unit/Integration Testing

Methodologies: Agile (Scrum), V-Model, Software Architecture, System Optimization

Certifications: AWS Academy Graduate, Google Cloud (AI Studio Prototypes)

LANGUAGES & INTERESTS

Languages: French (Native), English (Professional), Spanish (Intermediate)

Interests: Tech & AI Watch, Web3, Drawing, Running, Football, Hiking, Esport competitions